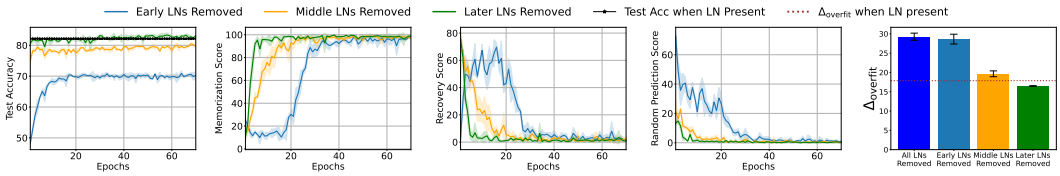
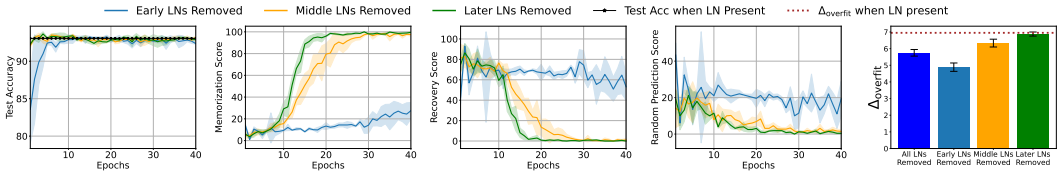


(a) Impact of early, middle, later LNs on learning (test) accuracy, memorization, recovery and random predicitons scores for Pre-LN models (DeiT, UTK-Face)



(b) Impact of early, middle, later LNs on learning (test) accuracy, memorization, recovery and random predicitons scores for Post-LN models (DeBERTa, Emotions)

Figure 3: **Pivotal impact of early LNs for learning and memorization across Pre- and Post-LN models.** (a) clearly shows impact of early LNs removal on destabilizing learning in Pre-LN models, accompanied with higher train-test-accuracy gap, $\Delta_{\text{overfit}}^{\text{Pre, early}}$, than later layers, whereas (b) shows early LNs removal help in suppressing memorization and improving recovery in Post-LN models, alongwith lower train-test-accuracy gap, $\Delta_{\text{overfit}}^{\text{Post, early}}$, than later layers.

In the Pre-LN model (DeiT, Fig. 3a), the removal of early LNs parameters significantly disrupts the learning process, highlighting their importance in learning for Pre-LN models. In the Post-LN model (DeBERTa, Fig. 3b), the removal of early LNs parameters mitigates memorization and enhances true label recovery most significantly compared to the cases of middle or later layers. This contrast highlights the pivotal impact of early LNs in shaping learning and memorization dynamics, positively in Post-LN and negatively in Pre-LN models. Aligned trends are observed for other multiple Pre- and Post-LN models, as presented in Appendix G.2. Prior studies [Gromov et al., 2024, Li et al., 2024, Lad et al., 2024, Men et al., 2024] highlighted the limited effectiveness of deeper layers for learning in Pre-LN models. Our observations take this a step further by precisely identifying that **LN in the early layers** is a critical factor in **memorization in Post-LN models**, presenting a *novel* and *distinctive* observation.

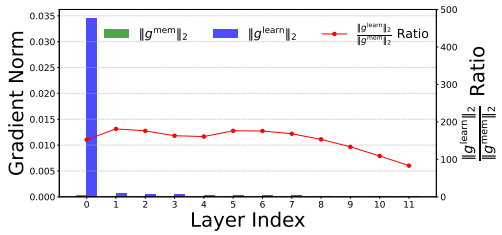
The distinctive effect of early LNs parameters removal—disrupting learning in Pre-LN models while mitigating memorization in Post-LN models—is further supported by the train-test accuracy gap (Δ_{overfit}). Specifically, in Pre-LN models, removing them leads to a more pronounced increase in $\Delta_{\text{overfit}}^{\text{Pre, early}}$ compared to middle or later LNs, whereas in Post-LN models, removing early LNs parameters results in a sharper decrease in $\Delta_{\text{overfit}}^{\text{Post, early}}$. This trend is shown in Fig. 3 (bar plots) and formalized as follows:

$$\Delta_{\text{overfit}}^{\text{Pre, early}} > \Delta_{\text{overfit}}^{\text{Pre, middle}} > \Delta_{\text{overfit}}^{\text{Pre, later}}, \quad \text{and} \quad \Delta_{\text{overfit}}^{\text{Post, early}} < \Delta_{\text{overfit}}^{\text{Post, middle}} < \Delta_{\text{overfit}}^{\text{Post, later}} \tag{3}$$

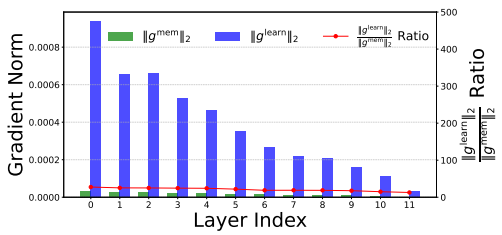
In summary, we observe that early layers LN are more significant than later layers LN, where their removal disrupts learning, explained by high $\Delta_{\text{overfit}}^{\text{Pre, early}}$ in Pre-LN models. On the other hand, their removal suppresses memorization while recovering true labels in Post-LN models, illustrated by low $\Delta_{\text{overfit}}^{\text{Post, early}}$. Similar trends are observed in other Pre-LN (GPTNeo, Qwen2, GPT2, ViT-B, ViT-S) and Post-LN (BERT, RoBERTa, ELECTRA, Longformer) models as illustrated in Appendix G.2.

6 Gradients Explain LN's Impact

To better understand the role of layer normalization (LN) in learning and memorization, we compute the gradient norms associated with both processes across different layers (g_x). Specifically, we measure the norms for learning (g_x^{learn}) and memorization (g_x^{mem}) gradients separately, allowing us to quantify their relative contributions throughout the network.



(a) Pre-LN (GPTNeo) $\left\|g_x^{\text{learn}}\right\|_2$ & $\|g_x^{\text{mem}}\|_2$ analysis across layers



(b) Post-LN (DeBERTa) $\left\|g_x^{\text{learn}}\right\|_2$ & $\|g_x^{\text{mem}}\|_2$ analysis across layers

Figure 4: **Learning vs. Memorization Gradients in Pre- and Post-LN Models:** (in Emotions Dataset) Results clearly exhibit high gradient norms of early layers LNs than later layers for both learning and memorization in Pre-LN (GPTNeo) and Post-LN (DeBERTa) models. Importantly, the learning gradient norm ($\left\|g_x^{\text{learn}}\right\|_2$) is consistently stronger than the memorization gradient norm ($\|g_x^{\text{mem}}\|_2$) across all layers. Furthermore, the ratio $\left\|g_x^{\text{learn}}\right\|_2 / \|g_x^{\text{mem}}\|_2$ is significantly higher in Pre-LN models compared to Post-LN models.

Theorem 1 (Learning Gradient Norm, $\|g_x^{\text{learn}}\|_2$ is greater than or equal to Memorization Gradient Norm, $\|g_x^{\text{mem}}\|_2$ across all layers). *It is formally represented as follows:*

$$\left\|g_x^{\text{learn}}\right\|_2 \geq \|g_x^{\text{mem}}\|_2 , \qquad \text{across all layers} \tag{4}$$

A proof of Theorem 1 is provided in Appendix B.

From Theorem 1, we observe that the learning gradient norms are generally greater than the memorizing gradient norms across all layers for both Pre- and Post-LN models. This observation is further validated empirically from the trend as seen in Fig. 4.

6.1 Understanding the Distinctive Impact of LN in Pre and Post-LN Architectures

Having identified the importance of early layers LNs (Sec. 5), we now focus on explaining why the removal of Layer Normalization (LN) in Pre-LN models hinders learning, while in Post-LN models, it mitigates memorization without disrupting learning. To do so, we focus on the ratio of learning-to-memorization gradients norms ($\frac{\|g_x^{\text{learn}}\|_2}{\|g_x^{\text{mem}}\|_2}$, red-color line plots in Fig. 4a & 4b) across layers. Based on the results, we uncover the following phenomenon:

$$\left.\frac{\|g_x^{\text{learn}}\|_2}{\|g_x^{\text{mem}}\|_2}\right|_{\text{Pre-LN}} \gg \left.\frac{\|g_x^{\text{learn}}\|_2}{\|g_x^{\text{mem}}\|_2}\right|_{\text{Post-LN}}, \qquad \text{across all layers} \tag{5}$$

This indicates that in Pre-LN models, LayerNorm primarily facilitates learning, as evidenced by the dominance of $\|g_x^{\text{learn}}\|_2$ over $\|g_x^{\text{mem}}\|_2$. Consequently, the removal of its parameters disrupts learning and exacerbates overfitting. In contrast, in Post-LN models, $\|g_x^{\text{learn}}\|_2$ and $\|g_x^{\text{mem}}\|_2$ are of comparable magnitudes. As a result, removing LN parameters effectively mitigates memorization by restoring genuine labels without disturbing learning. Consistent trends are observed for other Pre-LN (GPT2, Qwen2, ViT-B, DeiT, ViT-S) and Post-LN (RoBERTa, BERT, Longformer, ELECTRA) models, illustrated in Appendix G.3.

6.2 Why are LNs in Early Layers Important for Memorization and Learning?

In this section, we explain why the early layers LN are pivotal in governing memorization and learning across Post and Pre-LN models, through the lens of gradient analysis.

Theorem 2 (Gradient norm of loss $\mathcal{L}$ w.r.t input of LN is upper bounded).
Post-LN: Let z_i denote the input to LN$_1$ of the ith Post-LN model layer. Then,

$$\|g_{z_i}\|_2 = \left\|\frac{\partial \mathcal{L}}{\partial z_i}\right\|_2 \leq s_{\max}(P_1) \cdot \left(\frac{1}{\prod_{j=i}^{N} \left|1 - \sqrt{\text{Var}(\text{FFN}(x'_j))}\right| \left|1 - \sqrt{\text{Var}(\text{MHSA}(x_j))}\right|} \right) \cdot \cdot \prod_{j=i}^{N} \left(1 + s_{\max}(J_{\text{FFN}}^{x'_j})\right) \cdot \prod_{j=i+1}^{N} \left(1 + s_{\max}(J_{\text{MHSA}}^{x_j})\right) \tag{6}$$

Pre-LN: Let x_i denote the input to LN$_1$ of the ith Pre-LN model layer. Then,

$$\|g_{x_i}\|_2 = \left\|\frac{\partial \mathcal{L}}{\partial x_i}\right\|_2 \leq s_{\max}(P_2) \cdot \prod_{j=i}^{N} \left(1 + s_{\max}(J_{\text{FFN}}^{\text{LN}_2(x'_j)} J_{\text{LN}_2}^{x'_j})\right) \cdot \prod_{j=i}^{N} \left(1 + s_{\max}(J_{\text{MHSA}}^{\text{LN}_1(x_j)} J_{\text{LN}_1}^{x_j})\right) \tag{7}$$

A proof of Theorem 2, along with the expressions for LN$_2$ in both Pre- and Post-LN setup, is provided in Appendix C.

Theorem 3 (Upper bound of the gradient norm of Early Layers LN are higher than those of Later layers LN). *It is formally represented as follows:*

$$\text{UB}(\|g_{x_1}\|_2) \geq \text{UB}(\|g_{x_2}\|_2) \geq \cdots \geq \text{UB}(\|g_{x_N}\|_2) \text{ ; for both Pre- and Post-LN models} \tag{8}$$

where UB$(\|g_{x_i}\|_2)$ denotes the upper bound of $\|g_{x_i}\|_2$, and x_i is the input to the ith layer's LN.

A proof of Theorem 3 is provided in Appendix D.

The results depicted in Fig.4 empirically confirm the trend established in Theorem 3. Specifically, we observe that both $\|g_x^{\text{learn}}\|_2$ and $\|g_x^{\text{mem}}\|_2$ are significantly higher in the earlier layers compared to the later ones. This gradient decay trend is consistent across both Pre-LN (GPTNeo, Fig.4a) and Post-LN (DeBERTa, Fig.4b) architectures. Similar trends are observed for other Pre-LN (GPT2, Qwen2, ViT-B, DeiT, ViT-S) and Post-LN (RoBERTa, BERT, Longformer, ELECTRA) models, as shown in Appendix G.3.

Thus, the theoretical upper bounds not only provide an analytical explanation for the gradient magnitude behavior but also align closely with the empirical patterns observed across a wide range of transformer variants. This alignment helps explain why the removal of early layers LN parameters leads to disruption of learning in Pre-LN models and mitigation of memorization in Post-LN models, highlighting their predominant role in the entire network because of their higher gradient norm.

In addition to the isolation of learning and memorization in early layers, we observe another interesting pattern. For Post-LN models (Fig. 4b), both $\|g_x^{\text{learn}}\|_2$ and $\|g_x^{\text{mem}}\|_2$ decrease gradually over layers LN. However, for Pre-LN models (Fig. 4a), the gradient norms are predominantly high in the first layer, with the following layers having almost negligible norms. This observation explains why the removal of early layers LN parameters did not significantly affect learning for Post-LN models. That is because, in Post-LN models, later LNs can compensate for the absence of the early ones, recovering learning, while mitigating memorization, due to their comparable gradient norms. However, this does not hold for Pre-LN models, where the high gradient norms in the early layers LN are critical, and their absence severely disrupts learning.

Similar observations are observed in other Pre-LN (GPT2, Qwen2, ViT-B, DeiT, ViT-S) and Post-LN (RoBERTa, BERT, Longformer, ELECTRA) models, as shown in Appendix G.3.

In summary, gradient analysis highlights why removal of LN parameters significantly affects both learning and memorization: (1) **disrupts learning in Pre-LN models and mitigates memorization in Post-LN models** due to the distinct behavior of their gradient norms ratio; and (2) **it reveals the particular significance of early layer LNs**, which exhibit stronger gradient norms and thus play a more influential role in both learning and memorization processes.